

Output Form (OF)

Score: 3/5 — Moderate gaps | Confidence: high

Benchmark: MENTALCLOUDS | **Context:** Global South Mental Health Practitioners — India-Anchored OMHC / Peer-Support Deployment

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The benchmark output form (text-based structured summaries scored via ROUGE-1/2/L and BERTScore F1 [Q20, Q21, Q22, Q24] plus expert-rated clinical-acceptability and hallucination dimensions [Q44, Q46]) matches the deployment's text-summary modality. OF was rated LOWER priority and the modality match is strong. However, the metric form raises validity concerns: ROUGE F1 rewards lexical overlap [Q22, Q23] and is not weighted toward safety-critical content density — suicidal-ideation flags occupying small token spans contribute minimally to aggregate scores, regardless of clinical importance. The qualitative layer partially mitigates this but covers only three models on 39 test conversations [Q49, Q53, Q55], limiting statistical scope. Hallucination rates of 22–25% (i.e., 75–78% no-hallucination [Q55]) at a small evaluation scope represent a meaningful patient-safety signal that the form-level evaluation cannot fully resolve. A 2025 systematic review notes mental health AI benchmarks lack multi-session and safety-consistency evaluation [WEB-26]. The match is good at the modality level but the metric form imperfectly captures deployment-relevant signals.

ID	Evidence
Q20	"Given the generative nature of the task, we employ standard summarization evaluation metrics such as Rouge-1 (R-1), Rouge-2 (R-2), Rouge-L (R-L), and BERTScore (BS) along with their corresponding Precision (P), Recall (R) and F1 scores." (p.9)
Q21	"Since F1 accounts for Precision and Recall, we compare LLM's performance based on F1 unless stated otherwise." (p.9)
Q22	"ROUGE (Recall-Oriented Understudy for Gisting Evaluation) [58] assesses the overlap of n-grams (sequences of n consecutive words) between the generated summary and reference summaries." (p.9)
Q23	"This metric measures the number of overlapping units such as n-gram, word sequences, and word pairs between the generated summary evaluated against the gold summary typically created by humans." (p.9)
Q24	"ROUGE-L takes into account the longest co-occurring n-gram between the candidate and the reference summaries." (p.9)
Q44	"The evaluation framework encompasses six crucial parameters — affective attitude, burden, ethicality, coherence, opportunity costs, and perceived effectiveness." (p.12)
Q46	"Additionally, we incorporate a new parameter – the extent of hallucination. It is categorical – 0 (too much hallucinated), 1 (hallucination barely observed), and 2 (no hallucination observed)." (p.12)
Q49	"We select the three best LLMs (MentalLlama, Mistral and MentalBART) for the expert evaluation based on the automatic result." (p.13)
Q53	"Additionally, the evaluation of hallucination identification is divided into three categories: no hallucination observed, hallucination barely observed, and too much hallucinated in a set of 39 conversations." (p.14)
Q55	"Out of the 39 test conversations, the majority of cases on average demonstrate no hallucination, with Mistral and MentalBART achieving 75.13% and 76.15% each, while MentalLlama shows a slightly higher value of 77.69%." (p.14)
WEB-26	2025 systematic review: mental health AI benchmarks lack multi-session and safety-consistency evaluation (https://www.mdpi.com/2079-9292/15/3/524)

Strengths

- Output modality (structured text summaries) is well-matched to the deployment's text-out requirement [Q5, Q20].
- Dual quantitative + qualitative evaluation [Q5, Q18] provides complementary signal: lexical overlap plus expert acceptability and hallucination ratings [Q44, Q46].

- Per-component reporting [Q26, Q31, Q40] preserves granularity rather than collapsing to a single aggregate score.

ID	Evidence
Q5	"The generated summaries are evaluated quantitatively using standard summarization metrics and verified qualitatively by mental health professionals." (p.1)
Q18	"We undertake a comprehensive evaluation of the generated summaries across various architectures, employing a dual approach of quantitative and qualitative assessments." (p.9)
Q20	"Given the generative nature of the task, we employ standard summarization evaluation metrics such as Rouge-1 (R-1), Rouge-2 (R-2), Rouge-L (R-L), and BERTScore (BS) along with their corresponding Precision (P), Recall (R) and F1 scores." (p.9)
Q26	"Table 2 reports the automatic evaluation scores of LLMs on the summarization task for the Symptom and History (SH) psychotherapy element." (p.10)
Q31	"The experimental results presented in Table 3 focus on the summarization task for the Patient Discovery (PD) psychotherapy element." (p.10)
Q40	"Table 4. Results obtained on MentalCLOUDS for the summarization task on the Reflecting (RT) psychotherapy element." (p.11)
Q44	"The evaluation framework encompasses six crucial parameters — affective attitude, burden, ethicality, coherence, opportunity costs, and perceived effectiveness." (p.12)
Q46	"Additionally, we incorporate a new parameter – the extent of hallucination. It is categorical – 0 (too much hallucinated), 1 (hallucination barely observed), and 2 (no hallucination observed)." (p.12)

Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Yes — output modality (text summaries) matches deployment needs (text-in/text-out per deployment_context); both produce structured clinical summaries.

OF-2: Not applicable — no speech-based output is required for the deployment.

OF-3: Practitioner literacy is not a barrier (credentialed mental health professionals); accessibility requirements at the form level are met by text output.

OF-4: Modality match is strong; documented metric-form concern is that ROUGE F1 [Q21] and BERTScore reward lexical overlap rather than clinical-completeness or safety-critical recall — a partial external-validity concern. The qualitative layer is limited to 39 conversations and three models [Q49, Q55].

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Information Gaps

- Whether qualitative ratings included any weighting for safety-critical content recall.

Remediation

Gap 1: ROUGE F1 underweights safety-critical content; qualitative layer covers only three models on 39 conversations.

Recommendation: Introduce a content-coverage recall metric weighted by clinical importance (e.g., span-level risk-flag recall) and expand qualitative evaluation to a larger subset of conversations and models, including specific safety-content rubrics.